

Physics of billiard balls — Solution

X20 (2020) — English translation

A1

1.00 pts

Determine at what ratio between h and α after impact the ball will move without slipping.

The horizontal impulse imparted to the ball is equal to $\cos \alpha \int F(t) dt = mv$, where F is the impact force. On the other hand, the reported angular momentum about the center of mass is

$$\frac{2}{5}mR^2\omega = \left((h - R) \cos \alpha - \sqrt{R^2 - (h - R)^2} \sin \alpha \right) \int F(t) dt$$

In the case of отсутствия проскальзывания $v = \omega R$, так что we have coratio

$$\frac{2}{5} = \frac{h - R}{R} - \sqrt{1 - \frac{(h - R)^2}{R^2}} \operatorname{tg} \alpha$$

Answer: \textbf{Answer:}

$$\frac{2}{5} = \frac{h - R}{R} - \sqrt{1 - \frac{(h - R)^2}{R^2}} \operatorname{tg} \alpha$$

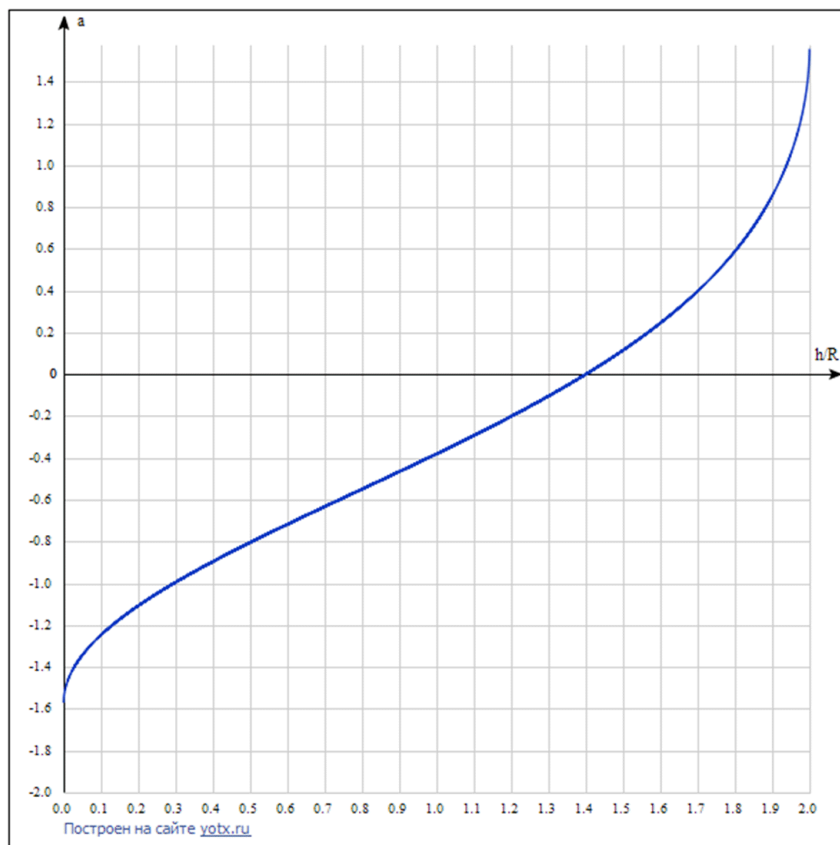
A2

0.50 pts

Draw a (qualitative) graph of $\alpha(h/R)$ showing the points corresponding to $\alpha = 0$ and $h = R$.

Answer: \textbf{Answer:}

$$\begin{aligned} \alpha = 0 &\Rightarrow h = \frac{7R}{5} \\ h = R &\Rightarrow \alpha = -\operatorname{arctg} \frac{2}{5} \end{aligned}$$



B1

1.00 pts

Find the speeds of the balls after expansion (modules and directions).

The force acting between the balls is directed at their point of contact along the line connecting the centers at the moment of contact, that is, at an angle $\phi = \arcsin \frac{b}{2R}$ to the direction of movement. It follows that the speed imparted to the second ball is also directed along this straight line. Then the ZSI together with the ZSE are given for the speeds of the balls: cue ball:

$$\vec{v}_c = \begin{pmatrix} -v \sin \phi \cos \phi \\ v \sin^2 \phi \end{pmatrix},$$

второй ball/sphere:

$$\vec{v}_o = \begin{pmatrix} v \sin \phi \cos \phi \\ v \cos^2 \phi \end{pmatrix}.$$

I.e. $|\vec{v}_c| = v \sin \phi$, $|\vec{v}_o| = v \cos \phi$

Answer: \textbf{Answer:}

$$|\vec{v}_c| = v \frac{b}{2R}, |\vec{v}_o| = v \sqrt{1 - \frac{b^2}{4R^2}}$$

B2

0.50 pts

What angle does the speed of the balls form after flying apart in such a model?

The speed of the 2 ball is directed at an angle ϕ to the original direction of movement, and the cue ball is directed at an angle $\pi/2 - \phi$. The total angle between the speeds of the balls is $\pi/2$.